

Localization

Multi-Object Disambiguation: front / back / left / right



Disambiguate by spatial relation and sort candidates

```

masks = sam3(rgb, "bowl") # returns ≥2 masks
candidates = [mask_to_world_point(m) for m in masks]

if "front" in instruction:
    candidates.sort(key=lambda p: p[0]) # sort by X
elif "back" in instruction:
    candidates.sort(key=lambda p: p[0], reverse=True)
elif "left" in instruction:
    candidates.sort(key=lambda p: p[1]) # sort by Y

target = candidates[0]
    
```

Motion Primitives

Linear Push: floor-plane primitive for flat objects



Linear push on floor plane

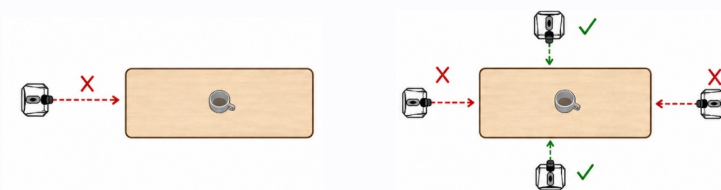
```

def push_object(obj_pos, target_xy):
    direction = (target_xy - obj_pos[:2]) /
    np.linalg.norm(target_xy - obj_pos[:2])
    pre_push = obj_pos[:2] - direction * 0.10
    post_push = target_xy

    move_to([pre_push[0], pre_push[1],
    obj_pos[2]])
    move_to([post_push[0], post_push[1],
    obj_pos[2]]) # Linear push
    
```

Navigation

Multi-Angle Approach: rotate approach vector



Sample rotated approaches for reachable pose

```

unit = (obj_pos[:2] - robot_pos[:2]) /
np.linalg.norm(obj_pos[:2] - robot_pos[:2])

for angle_offset in [0, np.pi/2, ...]:
    cos_off, sin_off = np.cos(angle_offset),
    np.sin(angle_offset)
    rotated_unit = np.array(...)
    target_xy = obj_pos[:2] - rotated_unit *
    approach_dist
    target_yaw = float(np.arctan2(...))
    if navigate_to_pose([target_xy[0], target_xy[1],
    target_yaw]):
        break
    
```

Skill Library (Partial)

• LOCALIZATION

Prompting strategy

Handle



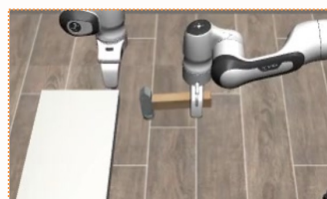
• MOTION PRIMITIVES

Pull

Turning

Handover

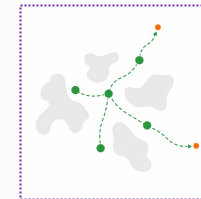
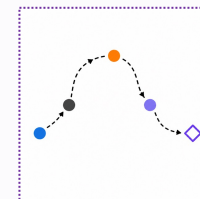
Push



• NAVIGATION

Way hopping

Free exploration



• OBJECT-LEVEL GRASPING

Wine

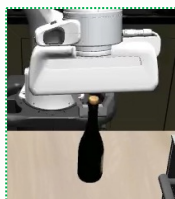
Bowl

Can

Pan handle

Box

Mug



• SCENE UNDERSTANDING

OBB axis

Minor vs Major

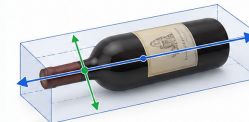
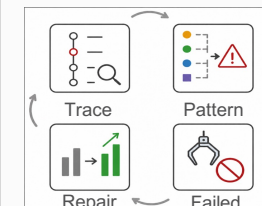


Table-aware

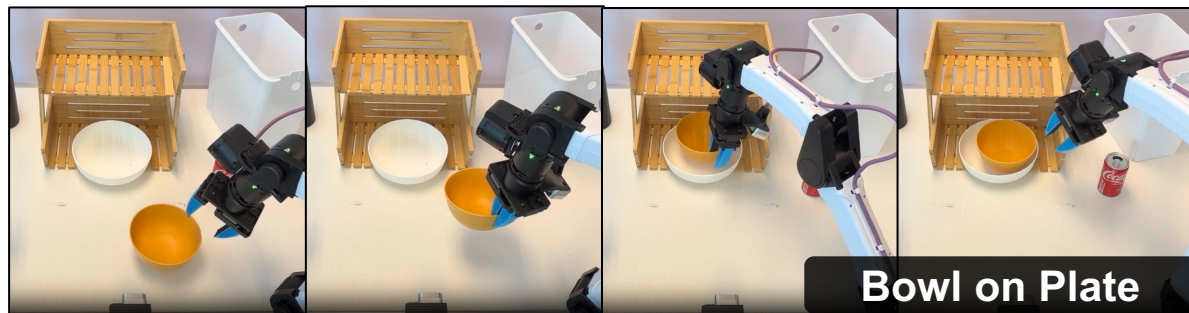


• DEBUGGING

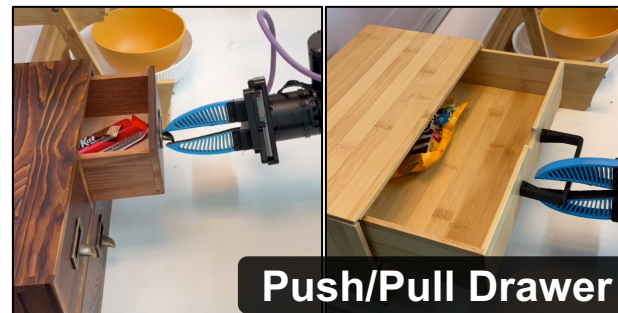
Workflow



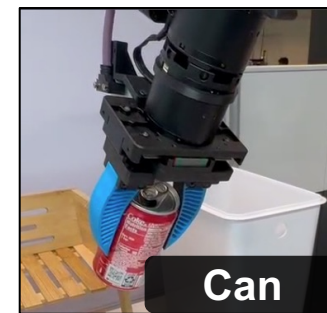
Simulation Skills Transfer to Real



Bowl on Plate



Push/Pull Drawer



Can



Handover